

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 315 — Past Questions & Materials
by Renance

Course: MEE 315
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

This study pack was compiled and restructured by Renance for MEE 315 students at Federal University of Petroleum Resources, Effurun. It brings together 2 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 2 — MEE 315 Fluid Mechanics II

Level 300 • First Semester • PastQuestion

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COLLEGE OF TECHNOLOGY & ENGINEFRING

DEPARTMENT OF MECHANICAL PNGINFERING

First Semester Examination 2024 20.

§ Fluid Mechanics I

Course & Title > MEE 34

Line Alfowed: 2 Hours

Instruction: Answer four (4) questions.

1. Given that, $U = \text{dax}(e = 3^\circ) V = \text{day}(an' = v7)$, e amine whether these velocity components represent a

physically possible two-dimensional flow, if so whether the flow 1s rotational or trretational.

b. With the aid of a simple sketch show a two-dimensional flow.

+2 Water flows from A to D and E through the series puis shown in below. Given the pipe diameters. velocities

and How rates below, solve for Q). v;. Qs. ds. Qs, Qy. and '5

v ve vow

QO, 7203

1.8m s7"

Q-?

B=?

dy = Sm

d,= 30mm

b. In practice. there will always be slight variations of velocity and pressure, hence, state four possible types

of flow,

3. Air flows between two parallel plates 80 mm apart. The following velocities were determined by direct

measurement.

['Distance from one plate (n (mm) io 10 [20 [30 [40 [50 [60 | 70 | 80

Velocity (mvs) : | 0 23 28 (31 32 29 22 14 0

Plot the velocity distribution curve and calculate the mean velocity.

b. Explain cavitation.

4. A veniure-meter having an inlet diameter of 75 mm and a throat diameter of 25 mm is used for measuring

the rate of flow of air through a pipe. Mercury U-tube gauges register pressures at the inlet and throat

equivalent to 250 mm and 150 mm of mercury, respectively. Determine the volume of air flowing through

the pipe per unit time in cubic metres per second. Assume adiabatic conditions ($\gamma = 1.3$). The density of air

at the inlet is 1.25 kg/m^3 and the barometric pressure is 760 mm of mercury. Take: Density of mercury 13.600

$\mu = 10^{-4} \text{ kg/m}\cdot\text{s}$

© State the factors that influence transition from laminar to turbulent boundary layer and at what Reynolds

Number does it occur. i

5. In the ideal flow around a half-body, the free stream velocity is 0.5 m/s and the strength of the source is $2.0 \text{ m}^2/\text{s}$

Determine the fluid velocity and its direction at a point. $r = 0.1 \text{ m}$ and $\theta = 120^\circ$.

b. What is vorticity and state three examples of application of vortex flow.

&. What are Turbomachines? Make a brief classification of Positive-Displacement Pumps (PDPs) designs.

b. What is Fanno flow. and what is supersonic flow.

Paper 2 of 2 — MEE 315 Fluid Mechanics II 2025

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERS

COLLEGE OF

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First Semestet

Course & Title: MEE 315 Fluid 'Mechanics

Time AHowed: 2 Hours

Instruction: Answer four (4) questions.

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* 2. Water flows from A to D and E through the series pipeline
and flow rates below, solve for Q Yb Q, da, Qs Q, and Xy

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O=? Gyn Bo ante

@, = 50mm B = c

es Pe wenidamimtrapesinde

Q='h =?

Bge?

d,=30mm

measurement.

f Distance from one plate (mm) 0 [10 [20-130 [40 [50]@O~ bo

| Velocity (m's) 0 23, | 28 |31 {32 | 29. eae)

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ie dn peacnde. there will always be slight variations of velocity and pressure. hence, state four possible losses

a8 Flow:

a) Spacing between ten parallel plates 50 mm apart. The following velocities were determined by direct PARR

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